Ysbyty Gwynedd
Anaesthetics Department
Adran Anaestheteg

PROVISION OF ANAESTHESIA IN REMOTE SITES **YG CG 901 v 3.0 NOV 2024.**

Emergency Anaesthesia in remote sites

(including the emergency department and critical care areas.)

REVIEW BY NOV 2027

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Key Points

1. All staff who work in ED, ITU, HDU or who work as part of a Cardiac Arrest, paediatric arrest or medical emergency team should be trained in basic airway management.
2. All staff who work in or attend ED, ITU or HDU & paediatric wards should regularly familiarise themselves with airway equipment available. This should be included in formal induction.
3. ED, ITU and HDU should at all times have availability of staff trained and skilled in rapid sequence induction and tracheal intubation and assistants skilled in assistance of such.
4. In all clinical locations where there is emergency airway equipment for adults or children, there must be evidence that it is regularly maintained, checked and ready to use.
5. Rapid sequence induction of anaesthesia and intubation outside of normal theatre working areas is a high-risk procedure. All intubators must be familiar with:
    a. the DAS guidelines for intubation of the critically ill adult,
    b. unanticipated failed intubation in RSI patients and
    c. NWTS guidelines for acute paediatric intubation.

    These are available on the DAS & NWTS websites.
6. The RSI checklist MUST be used in every intubation outside of a normal theatre working area where drugs are used for induction of anaesthesia (eg in ED, ICU, HDU, wards)
7. Waveform capnography must be attached and checked before every RSI and must be used at all times when patients are being intubated (including during cardiac arrest) where it is available. This includes ED, ITU, HDU, CCU and Dewi ward.
8. Waveform capnography must be used continuously in all intubated patients where available.
9. Waveform capnography must be attached when on inter-hospital or intra-hospital transfers of patients.

Introduction

- Airway interventions such as rapid sequence induction (RSI) in remote sites such as the ED are high-risk procedures with risk of causing severe harm or death to a patient.
- The key to minimising these risks is a well trained, well rehearsed team working together using a standardised approach and communicating their strategy.
- Refer to NAP4[1], the RCEM and RCoA joint position statement on Emergency Airway management[2] and the RCOA statement on anaesthetic services in remote sites[3].

Basic airway management & airway management in cardiac arrest.

1.1. Equipment for basic airway management should be available on every resuscitation trolley, and should be checked regularly as per SOPs.

1.2. All staff who respond as a team to attend cardiac arrests or medical emergencies, (including emergency department doctors and nurses) should be trained in basic airway management and follow national resuscitation guidelines.[4]

1.3. Intubation during cardiac arrest can be done (or supervised) by those who have had training in the necessary skills.

1.4. Waveform capnography should be used during cardiac arrest for all patients who have a supraglottic airway or are intubated.

Emergency anaesthesia (RSI) outside of theatres.

2. Rapid Sequence Induction is a high risk procedure outside of theatres and should only be done by those who have achieved at least initial assessment of competencies and have recent regular practice with RSI on patients.

2.1. A consultant anaesthetist should be called for all predicted difficult airways and all children who are being intubated in remote areas including the Emergency department.

2.2. Good preparation of the team, equipment and drugs is essential, as well as a solid plan consisting of a logical sequence of interventions in the event of failure.

2.3. **The RSI checklist <u>must</u> be used for all intubations outside of theatres / anaesthetic room when drugs will be given to assist intubation. [Appendix A]**

2.4. The plan for failure must be verbalised so that the whole team are aware of the plan and where the relevant equipment is if required.

2.5. In the event of a failed intubation follow the DAS guidelines.[5]

2.6. Waveform capnography must be used for all intubations and continued during transfer.

Airway equipment in the Emergency Department.

3. Airway equipment in ED is managed by the emergency department. All anaesthetists, ODPs and ED staff need to familiarise themselves with the following equipment. Any faults, missing, broken or soiled equipment should be reported immediately to the senior nurse on duty and a DATIX form completed.

3.1. Adult Airway Trolleys (x3): The airway trolleys contain all equipment needed for an adult intubation, including supraglottic airway devices, and cricothyroidotomy equipment. They should be restocked and sealed after each use with a tag. [Appendix B]



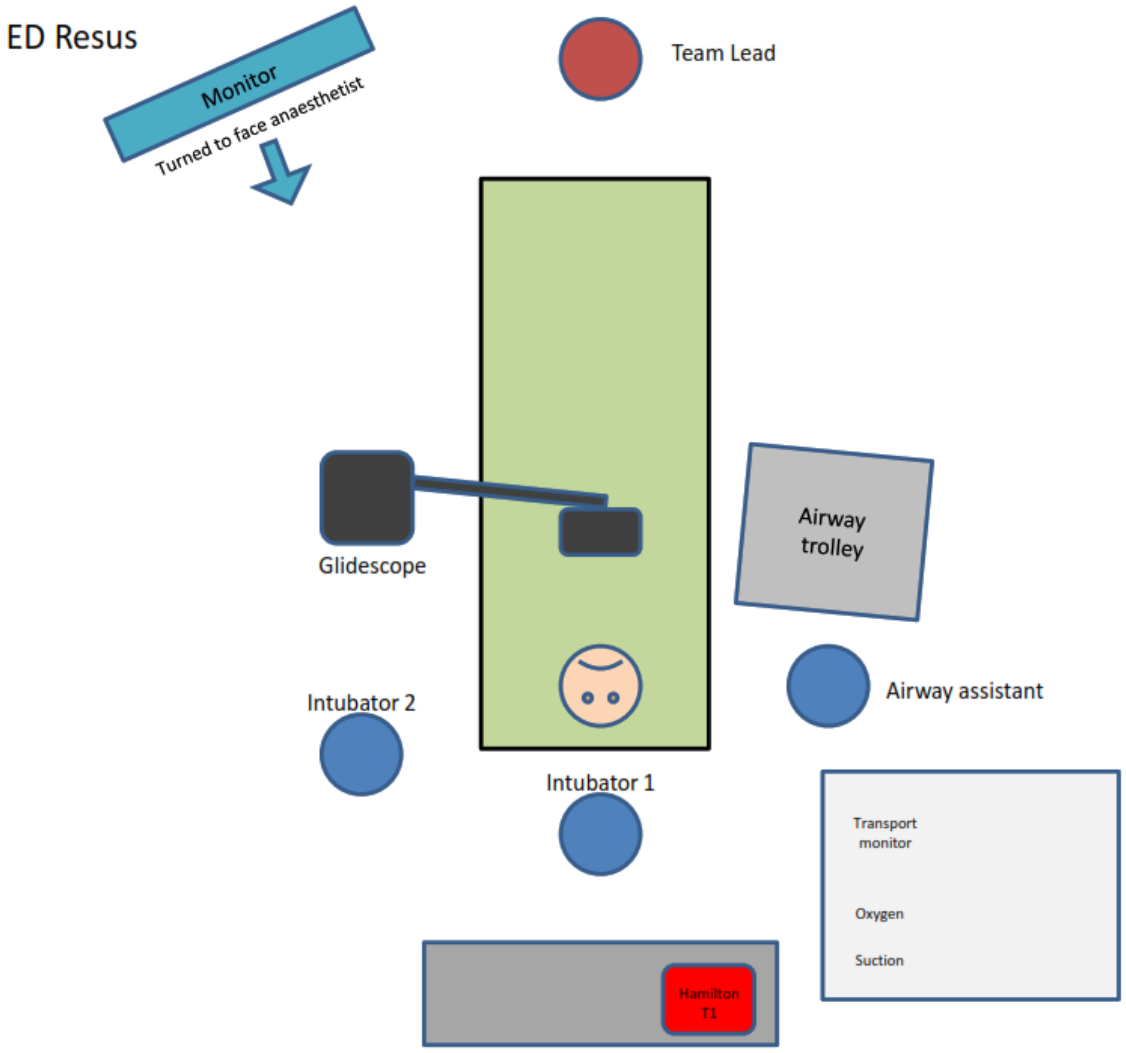
3.2. Paediatric Critical care trolley: The paediatric critical care trolley (in resus bay 1) has airway equipment for different age children from newborn to teenager, including a range of facemasks, supraglottic airway devices, cuffed and uncuffed tubes. It should be checked weekly and after every use. It should be sealed with a tag after every check.

3.3. **Ventilators.** There is a Hamilton T1 ventilator at each resus bay. The Hamilton should be checked daily by the ED nurses and after each use when a new disposable circuit is attached. An e-learning package is available online: https://www.hamilton-medical.com/en_GB/College.html .

3.4. **RSI Drug bags.** Drugs for RSI are kept in red bags in the controlled drug cupboard labelled ED: RSI 1 and ED:RSI 2. [Appendix C]



3.5. **Videolaryngoscope.** Always use the Glidescope as default unless there is concern about significant soiling of the airway. Make sure the airway assistant and second intubators can see the screen.
3.6. **Checklists.** Always use the pre and post RSI checklist. DAS guidelines for difficult and failed intubation are also available.
3.7. **Layout.** Use the following standard layout:



Theatres Airway Grab Bag:

4. On the shelf above the difficult airway trolley in theatres is a small portable airway grab bag which can be taken to remote areas such as the ward to assist in airway management (eg for cardiac arrests or RSI on the ward). The contents of this bag are checked weekly by the ODPs.

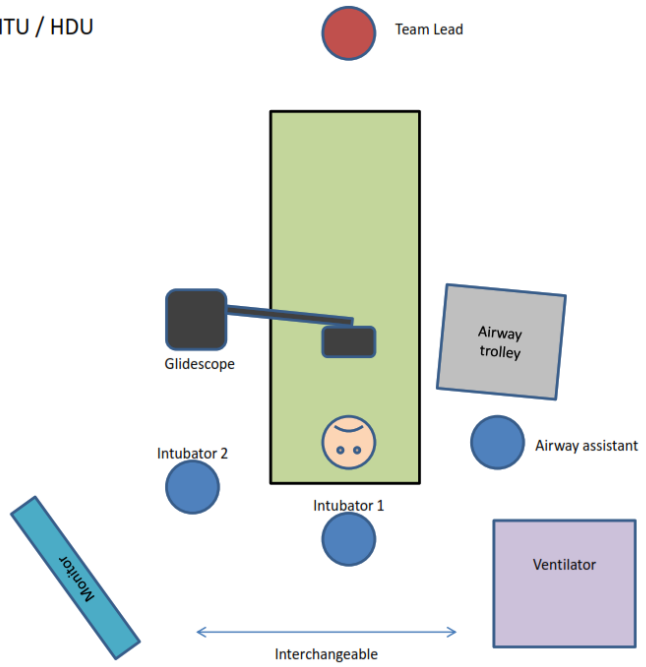
Airway equipment in ICU / HDU:

5. Airway equipment on ICU and HDU is managed by the ICU staff.
   5.1. On ITU and HDU every bed area should have an ambubag, waters circuit and a green tray containing laryngoscope, ET tubes and the RSI checklist.
   5.2. There is a Glidescope with disposable size 3, 4 and Low profile blades. The Glidescope should be your default first choice intubation equipment on ICU. Make sure the airway assistant and second intubators can see the screen.
   5.3. There is also an airway trolley for ITU and for HDU. The airway trolley will also have supraglottic airway devices, an airtraq disposable videolaryngoscope, a pentax videolaryngoscope and cricothyroidotomy (CICO) equipment.



   5.4. A single patient use flexible scope is also available (Ambuscope). Two sizes of scope are available. The thinner (1.8) version is for intubation and can be used with an Aintree catheter and via a supraglottic airway. The bigger (2.2) version is for bronchoscopy and has better suction capability.
   5.5. The RSI checklist and tracheostomy change checklists are on the difficult airway trolley along with the contents list.
   5.6. The ICU nurses are trained to assist with RSI. ODP's do not routinely attend ICU for intubations but can be called upon at the request of the intubator.
   5.7. Refer to DAS guidelines for tracheal intubation of critically ill adults.[6]

5.8. Layout. Use the standard layout:

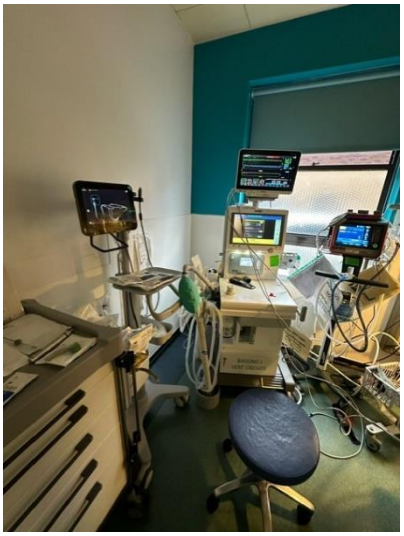


Airway equipment on Dewi (paediatric ward)

6. Equipment available in the "High Dependency" room on Dewi ward:
   6.1. A paediatric airway trolley stocked with paediatric basic and advanced airway equipment.
   - Paediatric airway guidelines including the NWTS guidelines and the YG Paediatric RSI checklist are printed and on the trolley. They are also available on the departmental guideline website.
   - Top drawer – Laryngoscopes and sundries
   - 2nd drawer – Facemasks, HME, Gueds
   - 3rd drawer – ETT tubes
   - This is checked and restocked by the ODPs. See photo



6.2. A Glidescope Videolaryngoscope with a range of disposable paediatric Mac, Miller and hyperangulated (LoPro) blades.



6.3. A Hamilton T1 ventilator. Use the neonatal circuit and expiratory valve for children <20kg. (see YG CG 807 for set up for children <20kg)

6.4. A Fabius Tiro anaesthetic machine. There is no scavenging and hence no vaporiser. A vaporiser can be requested from theatres in an emergency. If using volatile agent connect the scavenger to a Cardiff Aldosorber.

6.5. A cupboard with equipment for vascular access, monitoring, oxygen facemasks, nebulisers, BVM, invasive lines



6.6. RSI drugs are in the fridge in the little annexed store room within the HDU bay

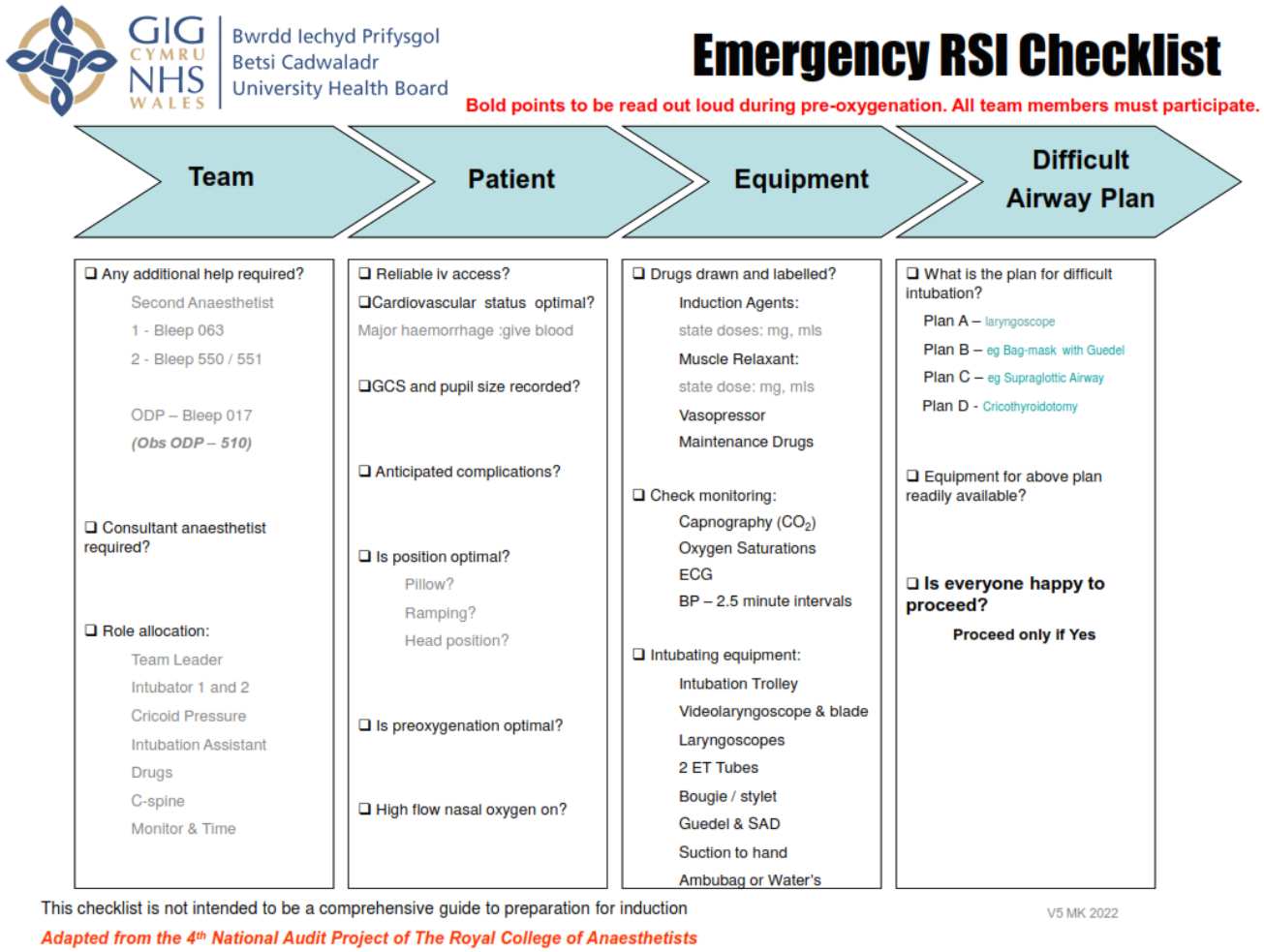


6.7. **Always call for senior help for a paediatric emergency intubation.**
6.8. **Refer to NWTS guidelines[7] for acute paediatric intubation.**
6.9. **Always use the Paediatric RSI checklist.[7]**
6.10. **Always use waveform capnography.**

References

1. NAP4 Royal College of Anaesthetists National Audit Project on complications of airway management.
2. RCEM and RCoA joint position statement on Emergency Airway management.
3. https://www.rcoa.ac.uk/system/files/RemoteSites2014.pdf
4. https://www.resus.org.uk/resuscitation-guidelines/adult-advanced-life-support/
5. https://das.uk.com/
6. https://das.uk.com/guidelines/icu_guidelines2017
7. https://www.networks.nhs.uk/nhs-networks/north-west-north-wales-paediatric-critical-care/documents/guidance-for-acute-paediatric-intubation/at_download/file

Appendix A –Pre and Post RSI checklists



GIG CYMRU NHS WALES
Bwrdd Iechyd Prifysgol
Betsi Cadwaladr
University Health Board

Emergency RSI Checklist

Bold points to be read out loud during pre-oxygenation. All team members must participate.

Team	Patient	Equipment	Difficult Airway Plan
❑ Any additional help required? Second Anaesthetist 1 - Bleep 063 2 - Bleep 550 / 551 ODP – Bleep 017 *(Obs ODP – 510)* ❑ Consultant anaesthetist required? ❑ Role allocation: Team Leader Intubator 1 and 2 Cricoid Pressure Intubation Assistant Drugs C-spine Monitor & Time	❑ Reliable iv access? ❑Cardiovascular status optimal? Major haemorrhage :give blood ❑GCS and pupil size recorded? ❑ Anticipated complications? ❑ Is position optimal? Pillow? Ramping? Head position? ❑ Is preoxygenation optimal? ❑ High flow nasal oxygen on?	❑ Drugs drawn and labelled? Induction Agents: state doses: mg, mls Muscle Relaxant: state dose: mg, mls Vasopressor Maintenance Drugs ❑ Check monitoring: Capnography (CO_2) Oxygen Saturations ECG BP – 2.5 minute intervals ❑ Intubating equipment: Intubation Trolley Videolaryngoscope & blade Laryngoscopes 2 ET Tubes Bougie / stylet Guedel & SAD Suction to hand Ambubag or Water's	❑ What is the plan for difficult intubation? Plan A – laryngoscope Plan B – eg Bag-mask with Guedel Plan C – eg Supraglottic Airway Plan D - Cricothyroidotomy ❑ Equipment for above plan readily available? **❑ Is everyone happy to proceed?** **Proceed only if Yes**

This checklist is not intended to be a comprehensive guide to preparation for induction

V5 MK 2022

Adapted from the 4th National Audit Project of The Royal College of Anaesthetists

GIG CYMRU NHS WALES
Bwrdd Iechyd Prifysgol Betsi Cadwaladr University Health Board

Post RSI & Pre Move Checklist

Bold points to be read out loud post Intubation. All team members must participate.

Airway & Breathing

❑ **ETCO2 trace?**

Target ETCO2 =

[3.5 to 4.0kPa for head injuries / intracranial bleed]

❑**Oxygen Saturations**

❑**Bilateral breath sounds**

Check tube length.

❑**ETT Length and Secure**

❑ **C-spine secure (trauma)**

Blocks and tape

❑**Sedation running**

If delayed consider rebolus of ketamine or midazolam

Circulation & Disability

❑**Target BP:**

MAP 85mmHg for neuro.

Penetrating trauma: titrate to radial pulse.

❑**At least 2 x iv (or io) access?**

Easy access to flush line.

❑**Pupil size checked & recorded**

❑ **Neuroprotective measures if indicated?**

See separate aide memoire

Equipment

❑ **Drugs to take?**

- Extra Muscle Relaxant
- Vasopressor
- Maintenance Drugs

❑ **Is all monitoring on?**

- Capnography (CO_2)
- Oxygen Saturations
- ECG (3 lead)
- BP 2.5 minute interval

❑ **Equipment to take:**

- Laryngoscope
- BVM
- Bougie & ET Tube
- Guedel & SAD
- Suction
- Drugs
- Oxygen & Ventilator
- Blood / Fluids

Packaging

❑ **Patient packaged on Wolverson mattress (trauma)**

❑ **What is the plan after CT?**

Consider calling EMRTS for transfer

❑**Lines accessible**

❑ **CT ready ?**

❑ **Team members required to prepare equipment post CT?**

Eg: Art line, Urinary catheter, syringe drivers, transfer trolley.

❑ **Happy to Proceed?**

This checklist is not intended to be a comprehensive guide to patient care.

V5 MK 2022

Appendix B –Emergency Department Airway Trolley

Airway Trolley Full Checklist

			Tick
Drawer 1 - Plan	Pentax videolaryngoscope & battery check		
3+1 attempts @ intubation	Laryngoscope blade Mac 4 and Handle, Catheter Mount and HME device		
	Tracheal tubes 7 & 8		
	Aquagel & Sleek 5cm		
	Nasal Specs		
Alternative laryngoscope	Glidescope blades Mac 3 & 4 & LoPro 4		
	Pentax disposable blade		
	Glidescope Stylet		
	Magil's Forcep		
Drawer 2 - Plan B			
3 attempts @ inserting Supraglottic	Ambu Auragain Laryngeal mask – sizes 3 & 4 & 5		
	iGel – sizes 3 & 4 & 5		
Drawer 3 - Plan C			
3 attempts @ Facemask ventilation with adjuncts	Nasopharyngeal Airways 6,7 & 8 (two of each)		
	Oropharyngeal Airway 2,3 & 4		
	Facemask 3,4 & 5		
Drawer 4 - Plan D			
Front of neck access Can't intubate, Can't oxygenate	Difficult Airway Society Front Of Neck Access kit		
	Melker Emergency Cricothyrotomy(1 Trolley only)		
	Chloraprep wand x2		
	Personnel eye protection		
Drawer 5 – Additional	Mapleson C (Water's) Circuit (2.0 litre bag)		
	MADgic Airway, Extra Portex Tracheal Tubes 6, 7 & 8 with Suction Reinforced Endotracheal tube 6, MLT 5 & 6 Catheter mount & HME		
	Adult BVM		
Side holder	Frova Bougie		
	Aintree Intubation catheter set		
	Bougie Ch12		
Ambu Scope	Ambu aScope Slim 3.8/1.2 x2		
	Monitor & power supply		
Difficult Airway Society Guidelines & Checklist			
	Print Name:		Sign:
All present, correct& functional. Trolley re-tagged.	Print Name:	Sign:	Date:

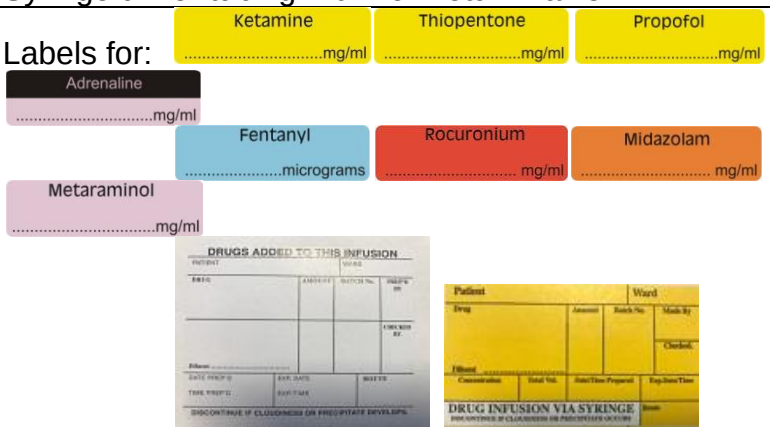
Version: Airway Trolley 2.3
Reviewed by PW, DM, MK 28/3/22

Author: Dr P. Williams YG-ED
Next Review Date: November 2022

Appendix C – Emergency Department RSI drug packs.

Ysbyty Gwynedd Emergency Department

Anaesthetic drug pack: Check list

		Tick
Anaesthetic Drugs		
1	Ketamine 200mg in 20ml	
1	Thiopental Sodium 500mg	
1	Propofol 1% (10mg/ml) 50ml	
1	Fentanyl 500mcg in 10ml	
2	Rocuronium 50mg in 5ml	
1	Midazolam 5mg in 5ml	
Emergency Drugs		
1	Metaraminol 10mg in 1ml	
1	Atropine 3mg in 10ml prefilled syringe	
1	Ephedrine 30mg in 10ml prefilled syringe	
1	Adrenaline 1mg in 10ml prefilled syringe	
2	Adrenaline 1mg in 1ml	
Sundries		
2	0.9% Saline 10ml prefilled syringe	
2	Water for injections 20ml (or 4 x 10ml)	
2	0.9% Saline 10ml amps	
1	100ml bag 0.9% Saline	
1	50ml syringe	
2	20ml syringe	
2	10ml syringe	
1	5ml syringe	
1	2ml syringe	
5	Drawing up filter needles	
1	Syringe driver tubing with non-return valve.	
2 each	Labels for: 	
Check complete		

Date	Name	Sign	Seal with red tag No:	Pack: 1 or 2 (delete as appropriate)

File completed checklists in back of resus room checks folder.

Ysbyty Gwynedd

Emergency Department

Anaesthetic drug pack: Restocking SOP

Standard Operating Procedure for the storage and record keeping of controlled drugs within the Emergency Department Anesthetic Drug pack

Anaesthetic drug pack contains the following Controlled Drugs (CD);

- Ketamine 200mg in 20 mL (x1)
- Midazolam 5mg in 5 mL (x1)
- Fentanyl 500mcg in 10 mL (x1)

The anaesthetic drug pack should be stored in the controlled drugs cupboard within Resus. Storage and records should be in line as per usual standard operating procedure (BCUHB Medicines Code).

Controlled drugs contained within the anaesthetic pack should be accounted for in the same running balance as all other controlled drugs within the cupboard. (There is no separate Controlled Drug register or order book for controlled drugs contained within the anesthetic drug pack).

Once a controlled drug from the anaesthetic drug pack has been administered a record should be made within the register and the pack replenished with ward stock.

When the anaesthetic pack is in use it should be under the direct supervision of a health care professional at all times.

Under no circumstances should the anaesthetic drug pack containing controlled drugs be removed from the department.